# Physical-Metric and Resolvent Certificates for the Weak-Disorder Anderson Construction

A Source-Specific Ram Calculus Audit of M523

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M523

October 7, 2026

Confidential report. Private corpus material remains local.
Catalogue/source pin: `adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

**Abstract**

We audit the complete supplied source of *Absolutely Continuous Spectrum for Weak-Disorder Anderson Models in Dimensions at Least Three*, separating finite Schur–Ward algebra, adaptive information requirements, and infinite-volume spectral arguments. We prove physical-coordinate residual bounds with explicit omitted-space and terminal-solve terms; a residual-to-Ward-form certificate controlling all neutral charges; a fixed-reference committed-state comparison avoiding multiplication of per-deletion tolerances; and an exact-range Gram clearing test. We establish an explicit weighted reset certificate and sharpen the classical Cayley readout, including a local reset witness outside the source's sufficient small-operator-norm hypothesis. The readout rank-error coefficient improves fourfold, and its transfer extends conditionally to bounded iid laws with bounded density. These results do not reproduce the all-scale probability proof, establish a new weak-disorder spectral theorem, compute an effective disorder threshold, or demonstrate a speedup. Contribution assessment: C3/5, the proved residual-to-form and fixed-reference certificates together with explicit local reset bounds give a notable verification extension while adaptive all-scale certificate existence and the headline spectral theorem remain unverified.

## Contents

**1 Pinned Problem and Scope** **2**
1.1 Source identity and reading boundary . . . 2
1.2 Operator, domain, law, and quantifiers . . . 3
1.3 The precise finite target . . . 3

**2 Stepwise Dependency Audit** **3**
2.1 State and initialization . . . 4
2.2 Ordinary updates and local trials . . . 4
2.3 Clearing and committed assembly . . . 5
2.4 Histories and spectral closure . . . 5

**3 Private Ram Correspondences** **6**

**4 Physical Resolvent Residual Certificates** **7**
4.1 Active dissipativity and terminal leakage . . . . . 7
4.2 Nonorthogonal synthesis and target correction . . . . . 7
4.3 Ward defects versus graph energies . . . . . 8

**5 Residual-to-Form and Committed-State Tests** **8**
5.1 Simultaneous control of every neutral charge . . . . . 8
5.2 Inexact harmonic elimination . . . . . 9

**6 A Certified Full-Range Clearing Test** **10**

**7 Explicit Weighted Reset and Form Bounds** **11**

**8 Sharp Classical Cayley Readout** **12**

**9 Conditional Infinite-Volume Extension** **14**

**10 Verification, Limits, and Research Directions** **15**
10.1 What a defensible finite record contains . . . . . 15
10.2 Verification scope . . . . . 16
10.3 Current primary literature and comparisons . . . . . 16
10.4 Pros, costs, and unmet dependencies . . . . . 17
10.5 Contribution assessment . . . . . 17

**A Read Scope and Evidence Locations** **17**

# 1 Pinned Problem and Scope

## 1.1 Source identity and reading boundary

Catalogue row 523 belongs to family 261. The source is dated September 23, 2026. Source references below mean the supplied local checkout at the stated pin, not a moving web version. Its directory is

```
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/openai-math/preprints/Absolutely-Continuous-Spectrum-for-Weak-Disorder-Anderson-Models-in-Dimensions-at-Least-Three-September-23-2026
```

All nine substantive files `build/01-introduction.tex` through `build/09-spectral.tex`, the driver `build/paper.tex`, the site-state figure, bibliography, and README were read. The driver includes no separate mathematical appendix: the long proofs embedded in Sections 3–9 were included. The supplied directory has the TeX sources but no local source `paper.pdf`. Reading is not independent verification of every inequality. Local results established in this report are distinguished from inspected but unresolved global dependencies.

### 1.2 Operator, domain, law, and quantifiers

For fixed integer $d \geq 3$, let

$$\Omega = [-1,1]^{\mathbb{Z}^d}, \qquad \mathbb{P} = \left(\tfrac{1}{2}\mathbf{1}_{[-1,1]}(v)\,dv\right)^{\otimes \mathbb{Z}^d}.$$

At fixed $\lambda > 0$, the source operator on complex $\ell^2(\mathbb{Z}^d)$ with counting measure is

$$(H_{\lambda,\omega}u)(x) = -\sum_{|e|_1=1} u(x+e) + \lambda\omega_x u(x). \tag{1}$$

The inner product $u^*v$ is linear in the second variable. $H$ is bounded self-adjoint on all of $\ell^2$, with $\|H\| \leq 2d + \lambda$; no boundary condition at infinity is selected. The site spectral measure $\mu_x(B) = \langle e_x, \chi_B(H)e_x\rangle$ has total mass one. For $z = E+i\eta$, $\eta > 0$, the convention $G(z) = (H-z)^{-1}$ gives $\operatorname{Im} G = \eta G^*G \succeq 0$.

The source's `thm:main` claims existence of one sufficiently small $e_2 \in (0, 1/100)$ and, for each fixed $d \geq 3$, $\lambda_d > 0$ such that for each fixed $0 < \lambda < \lambda_d$, almost surely

$$I_d \subset \sigma(H), \quad \chi_{I_d}(H)\ell^2 \subset \mathcal{H}_{\mathrm{ac}}(H), \qquad I_d = \begin{cases} (-6 + e_2/2, -6 + e_2), & d = 3, \\ -2d + (1/200, 1/100), & d \geq 4. \end{cases}$$

The interval is independent of $\lambda$; the probability-one event may depend on that fixed $\lambda$. There is no dimension-uniform assertion and no common event for every real coupling. The threshold for $d \geq 4$ uses an infinite family of finite-matrix probability conditions and is not an evaluated number. We neither establish nor refute the headline theorem.

### 1.3 The precise finite target

The scales are

$$\begin{aligned} &n_0 = \lfloor \log_4(3/\lambda^2) \rfloor, \quad m \geq 4n_0, \quad u = (\lambda^2/3)4^{n_0} \in (1/4, 1], \\ &s_n = u4^{-n}, \quad \eta = s_m, \quad t_n = s_n + \eta, \quad L = 4^{20m}, \quad N = L^d. \end{aligned} \tag{2}$$

$H_m$ is the periodic negative-hopping matrix on $(\mathbb{Z}/L\mathbb{Z})^d$ with independent true potentials, not a Dirichlet finite-volume operator. At each fixed $E \in I_d$, the claimed event has probability tending to one as $m \to \infty$, uniformly in $E$ for fixed $\lambda$. It supplies $A$, $|A| \geq N/2$, and $r$ with $|r_i| \leq C$ and $c \leq \operatorname{Im} r_i \leq C$, such that

$$\widehat{G} = (H_m - E - \eta \operatorname{diag}_A r)^{-1} \text{ exists}, \qquad \operatorname{Im} \widehat{G}_{ii} \geq c \quad (i \in A). \tag{3}$$

All real potentials, including on $A$, are the original true variables. Auxiliary randomness may construct a witness but cannot replace the true unconditional product law. A bounded set of tests at finite $m$ does not prove this quantified premise.

## 2 Stepwise Dependency Audit

Labels in this section are literal source labels. Their appearance records read dependencies, not independent certification of the multiscale proof.

### 2.1 State and initialization

Section 2, `prop:certificate`, states the finite target. True digits have $\lambda V_i = \sum_{k\ge 1} \lambda 2^{-k}\varepsilon_{i,k}$. An untouched suffix after $h$ digits is uniform on $[-\lambda 2^{-h}, \lambda 2^{-h}]$ and has variance $\lambda^2 4^{-h}/3$. The retained sigma-fields increase through numerical rollback. `lem:conditional-freshness` proves conditional product laws for unqueried digits and unused auxiliary variables at finite times and at almost surely finite stopping times by summing on $\{\tau = q\}$. This does not by itself prove that a selected pivot is unqueried. At fixed $E$, invertibility of every true principal block is a finite union of polynomial-null-set assertions, not a preliminary query of unrevealed potentials.

Section 3, `prop:initialization`, solves the scalar diagonal equation on the finite torus. Forward Fourier transforms are unnormalized; inverse transforms have factor $N^{-1}$ and continuum integrals $(2\pi)^{-d}dk$. The dispersion is $a(k) = -2d + 4\sum_j \sin^2(k_j/2)$. The two-resolvent estimate uses convex slices of dimension $d-1 \ge 2$, including shell tangencies. The smallest-grid-frequency gap is bounded by a directly counted positive patch, not an absolute quadrature error. It initializes $\|K\| \le 1/10$ and fourth-power rows with factor $1/10$ slack. The deliberately weaker estimate $\Lambda^+(y_\Sigma) \le C_d \|y\|_1^{4/3} \|y\|_2^{2/3}$ is retained in every $d \ge 3$. These Fourier and counting proofs were read; their smallness constants and interval thresholds were not computed.

### 2.2 Ordinary updates and local trials

Section 4 uses bit and reserve weights $w^{1/4}, \theta^{1/4}$ with $w + \theta = 1$, $\theta \in [1/4, 5/8]$; the final sweep has $\theta = 1/2$. The virtual port $g$ is a repeated and rescaled physical inverse, not an inverse on the duplicated space. The isometry for entrywise squares instead uses $k_x^2$, whose squared weights sum to one. These distinct maps must not be identified. The exact single-deletion formulas are

$$h' = h_{RR} - \gamma h_{Rx} h_{xR}, \qquad \gamma = \frac{\xi_0 + t r_x}{\sqrt{t}[1 + (\xi_0 + t r_x) g_{xx}]}.$$

Conditional centering, variance $t$, and $|\xi_0| \le C\sqrt{t}$ are inputs. The source derives $\mathbb{E}\gamma = -\sqrt{t}(g_{xx} - r_x) + O(t)$. `lem:single-update` gives $I - K' = \mathrm{Schur}_x(I - K) - J_x$ and $D' = \mathrm{Schur}_x D + L_x$. Corrections are padded onto one initial port space before summation. The complex symmetric $I - K$ is not Hermitian: its Schur contraction uses $\|K\| < 1$, not positivity. The Laplacian mean retains actual $p_i p_j$ weights in its Hilbert–Schmidt commutator. Fourth-row preservation uses the sharp inverse-form coefficient to get $(F^3)_{ii} \le C t^\kappa H_*$; Ward row sums alone are insufficient. The transformed-lag proof keeps survivor masks and does not factor dependent correction/forcing expectations.

Section 5, `lem:locality`, uses $d_T(i,j) \ge \frac{1}{2}\operatorname{dist}(i,j) - t^{-1.03}$. Its additive defect cannot be absorbed into a polynomial exponential prefactor. The proof splits distances at $t^{-2}$ before using rate $t^3$. `lem:inverse-row` combines short powers controlled by $Fp \simeq p$ and long powers controlled by $\|K\| < 1$, yielding an absolute inverse row bound $C\log(1/t)$. Tile side is $4^{6n} \asymp t^{-6}$, so test-space dimensions are polynomial in $t^{-1}$ for fixed $d$, not proportional to $N$. The stopped process includes its first violating update; all later increments are zero, with a predictable prestate indicator. Scalar variance envelopes permit the stated Golden–Thompson proof.

Budgets are $t^{.44\delta}$ for lag, $t^{.125\delta}$ for operator/form corrections, and $t^{.0005\delta}$ for normalized fourth-row proxies, with $\delta = .01$ and $\kappa = .005\delta$. Bounded overlap and exponential tails transfer local budgets to global caps. A first failure marks the entire tile pending, including unqueried sites. Retries use new auxiliary variables and orders. Numerical rollback does not restore true freshness. The final suffix sweep has no retry.

### 2.3 Clearing and committed assembly

Section 6, `lem:core-completion`, uses the symmetric unitary matrix $S = I + 2it\sqrt{p}\, g\sqrt{p}$ and the projection $\Pi$ onto $\operatorname{ran} S_{RB}$. The condition $\Pi_{ii} \le 2l_*$ gives a deterministic true-core completion bound through $\Delta S = \Pi \Delta S \overline{\Pi}$ and $\|\Delta S\| \le 2$, without controlling the Woodbury middle inverse. The conjugate on the right projector is essential. `lem:clearing-range-localization` localizes the entire range via the exterior equation and exponential conjugation, independently of its small singular values. Small raw row norms of $S_{RB}$ cannot replace that condition.

Preparation uses ridge leverages $m_i = [\mathcal{C}(\mathcal{C}^*\mathcal{C}+\tau I)^{-1}\mathcal{C}^*]_{ii}$. At a predictable maximizer, residual centering is a separate input. Unitary cancellation controls the fixed-ridge Gram increment; for $\mathcal{A} = \sum m_i$, $\mathcal{P} = \sum m_i^2$, the source proves $\mathbb{E}\Delta\mathcal{P} \le -(1 - o(1))m_*^2$ and $|\mathbb{E}\Delta\mathcal{A}| = o(m_*)$. The potential $\mathcal{P} - 10l_*\mathcal{A}$ also decreases when the ridge is lowered. Parameters are

$$\nu = 10^{-5}\delta/d, \quad l_* = t^{1+.44\delta}, \quad r_* = t^{-\nu}, \quad n_h = \lfloor t^{-1-.46\delta} \rfloor.$$

The heat limit needs $.46\delta - .44\delta - \nu > 0$. An individual leverage cap also needs a two-sided maximal estimate on the single varying-ridge stream. Provisional heat tests assume damping, lag, and row bounds, not a provisional $K$ or $D$ comparator. These probability and drift estimates remain read dependencies.

Section 7 freezes pending snapshot components at range $R_j = 4^{8j}$. Eligible cores have size at most $t_j^{-\nu}$ and inspection radius $4^{4j}/2$. Separation first localizes a prospective pivot and then proves it coupled and untouched by excluding other components and prior failed inspection neighborhoods. Thus freshness is not deduced from a matrix. The cutoff $j \le \lfloor m/2 \rfloor$ gives $\eta/t_j \le Ct_j$. Every unknown true core value is revealed and retained even though its completion bound is deterministic.

`prop:committed-assembly` transports forms only at committed endpoints. Its error has both changed surviving-edge energy and old energy through deleted sites. The latter is nonnegative and is subtracted in $L_a = D_a - \operatorname{Schur}_{O_a} D_{a-1}$. Halo overlap costs $t^{-d\nu}$; $.125\delta - d\nu > .10\delta$ closes the cumulative bound against one fixed initial form. Inverse comparison means quadratic forms on zero-extended neutral charges, not equality of full pseudoinverse matrices. `prop:reset` then closes a nonlinear contraction. Sections 5 and 7 below give local certificates, not replacements for the spatial and probabilistic estimates.

### 2.4 Histories and spectral closure

Section 8 orders all possible birth labels deterministically, with skipped and post-abort indicators zero. Successive conditioning gives product upper bounds for distinct prescribed labels, not independence. Witness extraction is followed by a union over deterministic witnesses; spatial separation prevents repeated leaf labels. The recursion uses $\alpha_{\mathrm{h}} = \nu/(100d)$ and pays the long initial nonrecursive range. Stretched exponentials compete against $N = 4^{20dm}$. Almost-sure retry termination remains separate from finite-history deterministic induction. `def:threshold` quantifies over all $m, n_0, u, E$ at individual probabilities; it is not effective computation.

Section 9 averages over an own-site variable with its complementary operator fixed, never over a law conditioned on adaptive success. A true spectral rank cut precedes active-space compression. Unconditional Wegner averaging bounds its expected rank. A bounded scalar test passes to zero height after a finite/infinite comparison with $\eta_m L_m \to \infty$. Ergodicity and rank-one changes propagate positivity. Purity still needs a complement-measurable exceptional energy set and a second own-site average. Sections 8–9 prove our conditional variant. No finite Schur identity establishes its certificate premise.

## 3 Private Ram Correspondences

These are actual passages inspected locally, not routing-map substitutes. No private text was used in public queries.

**R1: residual transport and checked reuse.** `C:/Users/mural/OneDrive/Documents/RamPatsCore50-TRs/TR-core-pat-5/TR-core-pat-5.tex`: Sections *Notation and Typed Graph Algebra*, *Coarsening, Lifts, and Schur Sources*, *Topology Edits and Checked Low-Rank Reuse*, *Residuals, Error Ledgers, and Query Certificates*; subsections *Polynomial evaluation without eigendecomposition* and *State machine and atomic commitment*. In particular `prop:schurresidual`, `eq:schurresidual`, `thm:woodbury`, `eq:filtererror`, and the inexact-action and adjoint residual formulas were inspected. For M523 they identify terminal leakage into active rows, current-version low-rank actions, and the separation of numerical state and retained observations. The Hermitian Chebyshev filter bound does not apply directly to a nonnormal damped inverse or complex entrywise-square $K$.

**R2: physical metric and omitted component.** `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01d-physical-metrics.tex` was read in full. Relevant headings are *A Nonorthogonal Representation Carries a Metric*, *The Full Residual Has an Omitted Component*, and *Approximating the Metric*; labels are `v21c01:physical-gram-definition`, `v21c01:metric-symmetry`, `v21c01:residual-decomposition`, `v21c01:residual-identity`, `v21c01:gram-equivalence`. They supply the Gram norm $c^*J^*Jc$, metric symmetry of compression, and the orthogonal full-residual decomposition. These matter for duplicated ports, localized tests, and nonorthogonal scattering-range coordinates. An exact compressed solve may have nonzero omitted residual.

**R3: no finite-to-infinite shortcut.** `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01f-spectral-limits.tex` was read in full, including `v21c01:pollution-proposition`, `v21c01:strong-not-norm`, `v21c01:stronger-contract`, *Domain-aware approximate inversion*, and *A Target-Specific Dual Estimate*. Its explicit graph-core example retains a spurious compressed eigenvalue despite strong resolvent convergence. The domain-aware inverse test needs both products on their actual spaces and a complementary-space argument. M523 is bounded, unlike that example, but fixed-height finite checks still do not identify spectral type at zero height.

**Attribution boundary.** Schur complements, Ward identities, Gram matrices, adjoint residual bounds, Cayley transforms, and spectral averaging are classical mathematics. The private Ram passages organize them as representation and verification contracts. Our contribution is their proved source-specific application with explicit failure conditions, not historical discovery of those techniques. No priority or patentability conclusion is made.

# 4 Physical Resolvent Residual Certificates

## 4.1 Active dissipativity and terminal leakage

Fix a finite real symmetric physical matrix $\mathcal{H}$, real energy $E$, active set $A \neq \varnothing$, and terminal set $T = A^c$. Terminal values are exact and $C_T = (\mathcal{H} - E)_{TT}$ is invertible. Write

$$B_{AT} = (\mathcal{H} - E)_{AT}, \qquad M = (\mathcal{H} - E)_{AA} - B_{AT} C_T^{-1} B_{TA},$$

and

$$B = M - t \operatorname{diag} r, \quad g = B^{-1}, \quad p_i = \operatorname{Im} r_i \geq p_0 > 0, \quad t > 0. \tag{4}$$

For empty $T$, all terminal terms below vanish. For every $v$, $-\operatorname{Im}(v^* B v) = t \sum_i p_i |v_i|^2$ implies $\|Bv\| \geq t p_0 \|v\|$. Hence

$$\|g\| \leq (t p_0)^{-1}. \tag{5}$$

This is an active-space bound. The full inverse damped only on $A$ need not satisfy the same estimate.

**Proposition 4.1** (Terminal-aware residual certificate)**.** *For an approximate full vector $y = (y_A, y_T)$ and right-hand side $f$, let $\mathcal{B} = \mathcal{H} - E - t \operatorname{diag}_A r$ and $e = f - \mathcal{B} y$. The exact active solution $x_A$ satisfies*

$$B(x_A - y_A) = e_A - B_{AT} C_T^{-1} e_T, \qquad \|x_A - y_A\| \leq \frac{\left\| e_A - B_{AT} C_T^{-1} e_T \right\|}{t p_0}. \tag{6}$$

*The terminal error is $x_T - y_T = C_T^{-1}(e_T - B_{TA}(x_A - y_A))$.*

*Proof.* Subtract the two block equations, solve the terminal error row, and substitute it into the active row. Apply (5). □

An approximate terminal solve must carry its actual transported residual or a validated bound on that action. A small active residual alone can miss a real terminal resonance. The source's almost-sure invertibility ensures existence, not a uniform numerical stability margin.

## 4.2 Nonorthogonal synthesis and target correction

**Theorem 4.2** (Physical residual and diagonal enclosure)**.** *Let $J : \mathbb{C}^k \to \mathbb{C}^A$ be injective, $G_J = J^* J > 0$, and $P_J = J G_J^{-1} J^*$. For $y = Jc$ approximating $gf$, set $e = f - By$ and $e_c = G_J^{-1} J^* e$. Then*

$$\|e\|^2 = \|e_c\|_{G_J}^2 + \|(I - P_J)e\|^2, \qquad \|gf - y\| \leq \|e\| / (t p_0). \tag{7}$$

*For a target $a$ and approximate dual vector $w$, put $d = a - B^* w$. Then*

$$|a^* g f - a^* y - w^* e| \leq \frac{\|d\| \, \|e\|}{t p_0}. \tag{8}$$

*For $a = f = e_i$, the imaginary part of $y_i + w^* e$ encloses $\operatorname{Im} g_{ii}$ with that radius. If $a_0 G_J \preceq \widehat{G}_J \preceq b_0 G_J$, $a_0 > 0$, then $\|e_c\|_{G_J} \leq a_0^{-1/2} \|e_c\|_{\widehat{G}_J}$.*

*Proof.* $P_J$ is the orthogonal projector onto $\operatorname{ran} J$ and $P_J e = J e_c$. Pythagoras proves the decomposition; $B(gf - y) = e$ proves the error bound. The identity $a^*(gf - y) = w^* e + d^*(gf - y)$ and Cauchy–Schwarz give the dual correction. Evaluate the lower Gram inequality at $e_c$ for the last assertion. □

The corrected target can be sharper when both residuals are small, but no performance improvement has been measured on the source tori. Residual evaluation, Gram enclosure, and evaluation of $w^*e$ need their own numerical error bounds. The perpendicular residual is mandatory even when an exact Galerkin solve gives $e_c = 0$.

### 4.3 Ward defects versus graph energies

**Proposition 4.3** (Inexact Ward identity)**.** *For any square candidate $Y$ and $R = I - BY$,*

$$\operatorname{Im} Y - tY^* \operatorname{diag}(p) Y = \frac{R^*Y - Y^*R}{2i}, \qquad \|\operatorname{Im} Y - tY^* \operatorname{diag}(p) Y\| \le \|R\| \, \|Y\| \,. \tag{9}$$

*Proof.* Multiply $B^* - B = 2it \operatorname{diag} p$ on the left by $Y^*$ and on the right by $Y$. Use $BY = I - R$ and $Y^*B^* = I - R^*$ and rearrange. □

For symmetric $Y$, define its approximate Laplacian from the nonnegative edge weights $tp_ip_j|Y_{ij}|^2$. Blindly substituting $\operatorname{Im} Y_{ii}$ into $\operatorname{diag}(p \operatorname{Im} Y_{ii}) - t \operatorname{diag}(p)|Y|^2 \operatorname{diag}(p)$ may not even annihilate constants, because (9) need not vanish. The exact symmetric inverse has both row and column Ward identities; an approximate inverse must not be assumed to have them.

## 5 Residual-to-Form and Committed-State Tests

### 5.1 Simultaneous control of every neutral charge

Let $\mathcal{N}_A = \{y \in \mathbb{C}^A : \sum_i y_i = 0\}$. Potentials modulo constants are identified with this Euclidean complement. For a symmetric matrix $h$ define

$$D_h(f) = \sum_{i<j} p_ip_j|h_{ij}|^2|f_i - f_j|^2.$$

This equals the source's half-sum over ordered pairs; diagonal entries of $h$ contribute no energy.

**Theorem 5.1** (Residual-to-Ward-form certificate)**.** *Let (4) hold, let $Y = Y^\top$, and put $\widehat{h} = \sqrt{t}\, Y$. Suppose certified column residuals obey $\|(I - BY)e_j\| \le \epsilon_j$. Define*

$$a_{ij} = \frac{\min(\epsilon_i, \epsilon_j)}{\sqrt{t}\, p_0}, \qquad L_a(f) = \sum_{i<j} p_ip_j a_{ij}^2 |f_i - f_j|^2. \tag{10}$$

*If $\widehat{D} = D_{\widehat{h}}$ has exactly the constant kernel and $L_a \preceq \kappa^2 \widehat{D}$ with $0 \le \kappa < 1$, then the exact source form $D = D_{\sqrt{t}g}$ has the same kernel and*

$$(1-\kappa)^2 \widehat{D} \preceq D \preceq (1+\kappa)^2 \widehat{D}, \tag{11}$$

$$(1+\kappa)^{-2} \widehat{D}^+ \preceq D^+ \preceq (1-\kappa)^{-2} \widehat{D}^+ \quad \text{on } \mathcal{N}_A. \tag{12}$$

*Writing $K_{ij} = tg_{ij}^2$ and $\widehat{K}_{ij} = \widehat{h}_{ij}^2$, one also has*

$$\left\| K - \widehat{K} \right\|_2 \le \max_i \sum_j a_{ij}(2|\widehat{h}_{ij}| + a_{ij}), \tag{13}$$

$$\left( \sum_j |h_{ij}|^4 \right)^{1/4} \le \left( \sum_j |\widehat{h}_{ij}|^4 \right)^{1/4} + \left( \sum_j a_{ij}^4 \right)^{1/4}. \tag{14}$$

*Proof.* $g - Y = g(I - BY)$ bounds every column-$j$ entry by $\epsilon_j/(tp_0)$. Symmetry gives the column-$i$ bound as well, proving $|h_{ij} - \widehat{h}_{ij}| \le a_{ij}$. On unordered edges set $(\mathcal{E}_h f)_{ij} = \sqrt{p_i p_j} h_{ij}(f_i - f_j)$. Its squared norm is $D_h(f)$, while $\left\|(\mathcal{E}_h - \mathcal{E}_{\widehat{h}})f\right\|^2 \le L_a(f)$. The triangle and reverse triangle inequalities give $(1-\kappa)\sqrt{\widehat{D}(f)} \le \sqrt{D(f)} \le (1+\kappa)\sqrt{\widehat{D}(f)}$. This identifies the kernel and the form inequalities. Inverse order on the positive neutral space proves (12). Factor the entrywise difference of squares to get (13); symmetry bounds both absolute row and column sums by its right side. Minkowski in $\ell^4$ proves (14). □

The theorem controls the source's distinct complex-square operator, fourth-power rows, and energy form. Small entry errors alone do not control relative energy at the smallest nonzero frequency. The hypothesis on $L_a$ explicitly retains that obstruction; it can be conservative and expensive. No volume-uniform cost or practical source-scale tolerance is claimed.

For any basis $Z : \mathbb{C}^{|A|-1} \to \mathcal{N}_A$ the relative test is exactly

$$K_D = Z^* \widehat{D} Z > 0, \quad K_a = Z^* L_a Z, \qquad \left\| K_D^{-1/2} K_a K_D^{-1/2} \right\| \le \kappa^2.$$

The Euclidean norm of $(Z^*Z)^{-1}K_a$ is not this relative norm. Nonsingular coordinate changes preserve the generalized inequality. Numerical eigenvalue enclosures used as rigorous certificates need validated arithmetic or independent analytic bounds, not just floating-point eigenvalues.

### 5.2 Inexact harmonic elimination

**Lemma 5.2** (Schur energy with an explicit solve defect)**.** *Let $D \succeq 0$ be a connected Laplacian, with nonempty survivors $R$ and eliminated $U \neq \varnothing$. Then $D_{UU} \succ 0$. For any $Z : \mathbb{C}^R \to \mathbb{C}^U$, let*

$$H_Z = \begin{pmatrix} I_R \\ Z \end{pmatrix}, \quad R_U = D_{UU} Z + D_{UR}, \quad S = \operatorname{Schur}_U D, \quad S_Z = H_Z^* D H_Z.$$

*Then*

$$S_Z = S + R_U^* D_{UU}^{-1} R_U. \tag{15}$$

*If $D_{UU} \succeq \beta_U I$, then $S_Z - \beta_U^{-1} R_U^* R_U \preceq S \preceq S_Z$.*

*Proof.* A zero-energy vector supported in $U$ would be globally constant and zero on $R$, hence zero. Complete the square with $Z = -D_{UU}^{-1} D_{UR} + D_{UU}^{-1} R_U$. Cross terms vanish and leave exactly (15); inverse order proves the enclosure. □

If $Z\mathbf{1}_R = \mathbf{1}_U$, the candidate preserves the constant mode. Without that identity $S_Z$ need not be a Laplacian although the formula remains valid. Its gauge must not be silently assumed correct.

**Theorem 5.3** (Fixed-reference committed-state enclosure)**.** *Let $D_0, D_k$ be exact connected Ward Laplacians on $A_0, A_k$, with $\varnothing \neq A_k \subset A_0$. Suppose verified candidates satisfy*

$$l_0 \widehat{D}_0 \preceq D_0 \preceq u_0 \widehat{D}_0, \qquad l_k \widehat{D}_k \preceq D_k \preceq u_k \widehat{D}_k, \qquad l_0, l_k > 0.$$

*Put $\widehat{S}_0 = \operatorname{Schur}_{A_0 \setminus A_k} \widehat{D}_0$. If $\mu \widehat{S}_0 \preceq \widehat{D}_k \preceq \nu_1 \widehat{S}_0$, $\mu > 0$, then for every neutral $y$ supported on $A_k$,*

$$\frac{l_0}{u_k \nu_1} D_0^+(y) \le D_k^+(y) \le \frac{u_0}{l_k \mu} D_0^+(y), \tag{16}$$

*where $y$ is extended by zero for $D_0^+$. No product indexed by earlier deletions occurs.*

*Proof.* Schur forms minimize energy over eliminated potentials. Minimization gives $l_0\widehat{S}_0 \preceq S_0 \preceq u_0\widehat{S}_0$, where $S_0 = \mathrm{Schur}_{A_0\setminus A_k} D_0$. Thus $(l_k\mu/u_0)S_0 \preceq D_k \preceq (u_k\nu_1/l_0)S_0$. Invert on the quotient. For a survivor-supported neutral charge,

$$D_0^+(y) = \sup_f\{2\Re(y^*f) - D_0(f)\} = S_0^+(y)$$

by minimizing first over the eliminated potential. This proves the claim. □

This is a finite verification extension of `prop:committed-assembly`: it compares an endpoint directly against the fixed stage reference instead of multiplying one numerical tolerance per operation. It does not assert that a test will pass or replace the source's spatial overlap proof. Lemma 5.2 supplies a certified Schur action if needed. A few sampled charges cannot prove the all-charge inequality. For $|A_k| = 1$ the neutral-space conclusion is vacuous.

# 6 A Certified Full-Range Clearing Test

The cross block $\mathcal{C} = S_{RB}$ can be small in amplitude yet carry a concentrated range direction. The following test can replace an exact zero-ridge limit when an exact range representation is established.

**Theorem 6.1** (Physical Gram certificate for the scattering range)**.** *Suppose $\mathcal{C} = ZW$ exactly, where $Z : \mathbb{C}^r \to \mathbb{C}^R$ is injective and $W : \mathbb{C}^B \to \mathbb{C}^r$ is onto. Set $G = Z^*Z$ and assume $aG \preceq \widehat{G} \preceq bG$, $a > 0$. For row $z_i$ define $\widehat{\ell}_i = z_i\widehat{G}^{-1}z_i^*$. Then*

$$\Pi_{\mathrm{ran}\,\mathcal{C}} = ZG^{-1}Z^*, \qquad a\widehat{\ell}_i \le \Pi_{ii} \le b\widehat{\ell}_i. \tag{17}$$

*In particular $b\widehat{\ell}_i \le 2t^{1+.44\delta}$ at every exterior site certifies the source's deterministic core-completion range condition. Rank zero is interpreted as $\Pi = 0$.*

*Proof.* Surjectivity gives $\mathrm{ran}\,\mathcal{C} = \mathrm{ran}\,Z$. The Gram projector is Hermitian, idempotent, and fixes $Z$. Inverse order gives $a\widehat{G}^{-1} \preceq G^{-1} \preceq b\widehat{G}^{-1}$. Evaluate at $z_i^*$. □

No uniform lower bound on nonzero singular values of $\mathcal{C}$ is imposed. Instead, exact range equality and the rank of the representation must be proved. A well-scaled $Z$ may place tiny amplitudes in $W$; verifying surjectivity of $W$ remains necessary and may be difficult.

**Proposition 6.2** (An approximate range is insufficient)**.** *No bound tending to zero with $\left\|\mathcal{C} - \widetilde{\mathcal{C}}\right\|$ alone controls $\left\|\Pi_{\mathrm{ran}\,\mathcal{C}} - \Pi_{\mathrm{ran}\,\widetilde{\mathcal{C}}}\right\|$. Small ridge leverage alone cannot certify the full-range condition.*

*Proof.* Let $\mathcal{C}_\epsilon = \mathrm{diag}(1, \epsilon)$, $\widetilde{\mathcal{C}} = \mathrm{diag}(1, 0)$, $\epsilon > 0$. Their distance is $\epsilon$, while the range projectors differ by $\mathrm{diag}(0, 1)$ and have distance one. At ridge $\tau > 0$, the second leverage is $\epsilon^2/(\epsilon^2 + \tau)$, arbitrarily small, but the exact projector diagonal remains one. □

This is an information obstruction, not a counterexample to the source range-localization lemma. An unqualified truncated SVD changes its target. Once exact range bounds hold, the source's unitary proof yields $\Delta S = \Pi\Delta S\overline{\Pi}$, $\|\Delta S\| \le 2$, and $|\Delta S_{ij}| \le 2\sqrt{\Pi_{ii}\Pi_{jj}}$. The relation $\Delta S = 2i\sqrt{t}\,\sqrt{p_R}\Delta h\sqrt{p_R}$ then gives the entry and row bounds with their physical weights. This remains compatible with large intermediate Woodbury coefficients; those coefficients provide no hidden numerical stability guarantee.

# 7 Explicit Weighted Reset and Form Bounds

This refinement of source `prop:reset` uses the physical damping metric to normalize the inverse. Its derivation and independent checks are recorded in the audit. The improvement is a finite-state certificate with explicit constants; it does not improve an all-scale exponent or probability estimate.

**Theorem 7.1** (Weighted radii certificate)**.** *Let $M = M^\top$ be finite real symmetric, $t > 0$, $r \in \mathbb{C}^A$ with every $p_i = \operatorname{Im} r_i > 0$, and $g = (M - t \operatorname{diag} r)^{-1}$. Put $q_i = \operatorname{Im} g_{ii}$, $D_p = \operatorname{diag} p$, $P = \operatorname{diag}(\sqrt{p})$, and*

$$K_{ij} = t g_{ij}^2, \quad K_w = D_p^{-1} K D_p, \quad \beta = \max_i q_i/p_i, \quad e = \left\| D_p^{-1}(\operatorname{diag} g - r) \right\|_\infty .$$

*Assume $I - K_w$ invertible and $A_\infty \geq \left\|(I - K_w)^{-1}\right\|_{\infty\to\infty}$ is established. If $0 < \delta < 1$ satisfies*

$$A_\infty \left( e + \frac{\beta \delta^2}{1 - \delta} \right) \leq \delta, \qquad A_\infty \beta \frac{\delta(2 - \delta)}{(1 - \delta)^2} < 1, \tag{18}$$

*then there is a unique $z$ with $\max_i |z_i|/p_i \leq \delta$ such that*

$$\operatorname{diag}\left((M - t \operatorname{diag}(r + z))^{-1}\right) = r + z.$$

*The new damping satisfies $p_i' = \operatorname{Im}(r_i + z_i) \geq (1 - \delta) p_i$. No uniform ratio $\max p / \min p$ or condition $\|K\|_2 < 1/2$ is required for this finite result.*

*Proof.* Normalize the inverse by

$$Q = tPgP = (B_0 - iI)^{-1}, \qquad B_0 = t^{-1} P^{-1}(M - t \operatorname{diag}(\Re r)) P^{-1}.$$

$B_0$ is real symmetric, so $Q$ is a symmetric normal contraction and $QQ^* = \operatorname{Im} Q$. In particular $\|Q_{i\cdot}\|_2^2 = t p_i q_i$. Write $z = D_p w$, $\Delta = \operatorname{diag} w$. For $\|w\|_\infty \leq \delta < 1$,

$$g_z = t^{-1} P^{-1} Q (I - \Delta Q)^{-1} P^{-1}.$$

In the diagonal term of degree $k \geq 1$, use the two endpoint rows of the symmetric $Q$, the $k$ factors of $\Delta$, and the remaining contraction factors. Its modulus is bounded by $(tp_i)^{-1}(tp_i q_i)\delta^k = q_i \delta^k$. The linear term is exactly $(Kz)_i$. Thus the remainder $\mathcal{R}(z) = \operatorname{diag}(g_z) - \operatorname{diag}(g) - Kz$ satisfies

$$\left\| D_p^{-1} \mathcal{R}(D_p w) \right\|_\infty \leq \frac{\beta \delta^2}{1 - \delta}.$$

For two $w$ in the ball, telescope each $k$-linear product. The Lipschitz constant for the remainder is at most $\beta \sum_{k \geq 2} k \delta^{k-1} = \beta \delta (2 - \delta)/(1 - \delta)^2$. The map

$$w \longmapsto (I - K_w)^{-1} \left[ D_p^{-1}(\operatorname{diag} g - r) + D_p^{-1} \mathcal{R}(D_p w) \right]$$

preserves the ball and is contractive by (18). Banach's theorem proves existence and uniqueness there. The damping bound follows from $|\operatorname{Im} z_i| \leq \delta p_i$. Iteration from zero also proves measurability in the finite input data. □

The inverse row norm is a real obligation. A verified approximate inverse $C$ with $\|I - (I - K_w)C\|_\infty < 1$ provides an inverse bound by a Neumann series in finite dimension. A spectral-radius observation does not provide the required induced norm bound. In the source, the logarithmic row estimate comes from locality and Ward structure and remains a global dependency.

**Proposition 7.2** (Explicit weighted Ward-form transport)**.** *Under the preceding hypotheses, for any perturbation with* $\max_i |z_i|/p_i \le \delta < 1$*, let* $D, D'$ *be the old and new Ward forms, using* $p, p'$ *respectively. Then*

$$\frac{(1-\delta)^2}{(1+\delta)^4} D \preceq D' \preceq \frac{(1+\delta)^2}{(1-\delta)^4} D. \tag{19}$$

*Their kernels agree. On a common neutral space inverse factors are reciprocated in reversed order; further Schur elimination preserves these form factors. If* $z$ *is real, they improve to* $(1+\delta)^{-4}$ *and* $(1-\delta)^{-4}$*.*

*Proof.* For real diagonal multiplication $f$, $h = \sqrt{t}\, g$ and $h' = \sqrt{t}\, g_z$, the resolvent commutator identity gives

$$P[f, h']P = (I - Q\Delta)^{-1} P[f, h]P (I - \Delta Q)^{-1}.$$

It follows either by taking commutators of $h'^{-1} = h^{-1} - \sqrt{t}\,\mathrm{diag}\, z$ or multiplying out; $f$ commutes with the diagonal perturbation. Both outer factors have norm at most $(1-\delta)^{-1}$; their inverses have norm at most $1+\delta$. Changing from $p$ to $p'$ multiplies on both sides by $\mathrm{diag}\,\sqrt{p'/p}$, with singular values between $\sqrt{1-\delta}$ and $\sqrt{1+\delta}$. Use $2D(f) = \|P[f,h]P\|_{\mathrm{HS}}^2$. This yields (19); complex potentials follow by separating real and imaginary parts in the real symmetric graph forms. Positive factors preserve the kernel. Inversion on the quotient reverses order, and Schur minimization preserves order. For real $z$, $p' = p$, so the extra diagonal factors disappear. □

**Exact witness outside the source's sufficient norm restriction.** Take $M = 0$, $t = 1$, $r = i\,101/100$, and $\delta = 1/50$. Then

$$A_\infty = \frac{10201}{20201}, \quad \beta = \frac{10000}{10201}, \quad e = \frac{201}{10201},$$

and the two left sides of (18) are, respectively,

$$\frac{10049}{989849} < \frac{1}{50}, \qquad \frac{990000}{48502601} < 1.$$

The exact corrected solution $r' = i$ has $|r' - r|/p = 1/101 < 1/50$. Here $\|K\|_2 = \beta > 1/2$. Thus the local inverse-based sufficient criterion genuinely admits a state outside the source reset proposition's $\|K\| < 1/2$ hypothesis. It is a one-site witness, not an Anderson stage satisfying every source invariant; its neutral energy space is trivial. Nontrivial two- and three-site form checks are separately recorded by the parent. The example does not weaken the small-$K$ assumptions elsewhere in the source induction.

## 8 Sharp Classical Cayley Readout

The following sharpening uses elementary classical Cayley algebra; it is not a new transform theorem.

**Lemma 8.1** (Sharp diameter and rank)**.** *For finite self-adjoint* $B, C$ *and* $\eta > 0$*, set* $f_\eta(B) = B(I - i\eta B)^{-1}$*. Then*

$$f_\eta(B) - f_\eta(C) = (I - i\eta B)^{-1}(B - C)(I - i\eta C)^{-1}, \tag{20}$$

$$\|f_\eta(B) - f_\eta(C)\| \le \eta^{-1}, \qquad \|f_\eta(B) - f_\eta(C)\|_{\mathrm{HS}}^2 \le \eta^{-2}\,\mathrm{rank}(B - C). \tag{21}$$

*The norm constant is sharp.*

*Proof.* Multiplication on the left and right by the denominators cancels the mixed product $-i\eta BC$, proving (20) without commutation. It bounds the rank by $\operatorname{rank}(B-C)$. The matrix $U_B = (I+i\eta B)(I-i\eta B)^{-1}$ is unitary and $f_\eta(B) = (U_B - I)/(2i\eta)$. The difference norm is at most $1/\eta$; its squared Hilbert–Schmidt norm is at most rank times squared norm. Scalars $B = 1/\eta, C = -1/\eta$ attain it. □

**Theorem 8.2** (Generalized finite readout)**.** *Let $H_N$ be a finite real symmetric lattice Anderson matrix with independent site densities bounded by $\rho_\infty$. Fix real $E$, $\eta > 0$, $\alpha, \pi_0 \in (0,1]$, and positive $B_1, p_-, p_+, q_*$. Suppose with unconditional probability at least $\pi_0$ there exist $A$, $|A| \geq \alpha N$, and $r$ with $|r_i| \leq B_1$, $p_- \leq \operatorname{Im} r_i \leq p_+$ such that the inverse in (3) exists and has active diagonal imaginary parts at least $q_*$. Put*

$$C_R = 1 + (B_1+1)/p_-, \quad c_* = q_*/(p_+ C_R^2), \quad M_\epsilon = 1 + \epsilon^{-2}, \quad \Phi_\theta(v) = \min\{(1+\theta)M_\epsilon v, c_*\}$$

*for $0 < \epsilon \leq 1$, $\theta > 0$. With $q_i = \operatorname{Im}(H_N - E - i\eta)^{-1}_{ii}$,*

$$\mathbb{E}\frac{1}{N}\sum_i \Phi_\theta(q_i) \geq \alpha\pi_0 c_* - 2\rho_\infty \epsilon(1 + 1/\theta). \tag{22}$$

*No independence of $A, r$ from the true variables and no measurable selection of a certificate are assumed.*

*Proof.* The feasibility event is Borel: for each $A$, constrain $r$ and potential values to compact boxes and impose a determinant lower bound $1/j$. Inverse inequalities are closed there and their projection is compact. Take a countable union over $j$ and boxes and a finite union over $A$. All real principal blocks at fixed $E$ are invertible almost surely, since their determinants have a nonzero highest diagonal monomial and the joint potential law has a density.

For a qualifying witness, let $S_A$ be the real Schur complement after eliminating $A^c$, $\widehat{G}_A = (S_A - \eta \operatorname{diag} r)^{-1}$, and $G_A = (S_A - i\eta I)^{-1}$. Resolvent identity and the active dissipative bound give $\left\|\widehat{G}_A e_i\right\| \leq C_R \|G_A e_i\|$. Ward gives $q_* \leq \eta p_+ \left\|\widehat{G}_A e_i\right\|^2$, so $\eta \|G_A e_i\|^2 \geq c_*$. Let $T = (H_N - E)^{-1}$ and compress to $A$: $B_A = P_A T P_A$, hence $G_A = f_\eta(B_A)$. Define

$$Q = \chi_{[E-\epsilon\eta, E+\epsilon\eta]}(H_N), \quad r_Q = \operatorname{rank} Q, \quad T_c = T(I-Q), \quad B_c = P_A T_c P_A, \quad G_c = f_\eta(B_c).$$

Both compressions are self-adjoint, without $P_A Q = Q P_A$. Lemma 8.1 gives $\|D_c\|_{\mathrm{HS}}^2 \leq r_Q/\eta^2$ for $D_c = G_A - G_c$. Functional calculus for $B_c$ and the scalar inequality outside the cut give

$$\eta \|G_c e_i\|^2 \leq \eta \|B_c e_i\|^2 \leq \eta \|T_c e_i\|^2 \leq M_\epsilon q_i.$$

Weighted squared triangle inequality implies $c_* \leq (1+\theta)M_\epsilon q_i + (1+1/\theta)\eta \|D_c e_i\|^2$ for $i \in A$. Truncate the first term at $c_*$ and sum to obtain

$$N^{-1}\sum_i \Phi_\theta(q_i) \geq \alpha c_* \mathbf{1}_{\text{feasible}} - (1+1/\theta) r_Q/(\eta N).$$

The inequality also holds off feasibility, because its left side is nonnegative there.

For own-site averaging, freeze all other potentials. The site diagonal resolvent is $1/(s - M(z))$, $\operatorname{Im} M(z) > 0$. Its imaginary part integrates to $\pi$ over $s \in \mathbb{R}$. Multiply by the bounded density and integrate against a continuous nonnegative energy test; the Poisson approximate identity yields $\mathbb{E}_s \mu_{i,s}(J) \leq \rho_\infty |J|$ for every Borel $J$. Summing sites at $J = [E - \epsilon\eta, E + \epsilon\eta]$ gives $\mathbb{E} r_Q \leq 2\rho_\infty \epsilon\eta N$. Take unconditional expectation in the previous pointwise bound. □

The choice

$$\epsilon \le \min\left\{1, \frac{\alpha\pi_0 c_*}{4\rho_\infty(1+1/\theta)}\right\}$$

leaves lower expectation $\alpha\pi_0 c_*/2$. For $\alpha = \pi_0 = 1/2$, $\rho_\infty = 1/(2\lambda)$, $\theta = 1$, this is

$$\epsilon \le \min\{1, c_*\lambda/16\}, \qquad \mathbb{E}N^{-1}\sum_i \min\{2M_\epsilon q_i, c_*\} \ge c_*/8. \tag{23}$$

The source uses $c_*\lambda/64$. Its rank penalty $8r_Q/(\eta N)$ becomes $2r_Q/(\eta N)$, since the squared norm improves by four. This permits a smaller truncation slope for the same expectation; it does not improve the disorder threshold or prove certificate existence.

# 9 Conditional Infinite-Volume Extension

**Theorem 9.1** (Bounded-density transfer conditional on certificates)**.** *On $\ell^2(\mathbb{Z}^d)$ let hopping be as in (1) and diagonal values iid with a bounded density $\rho$ supported in a fixed compact interval. Suppose on a nonempty bounded open interval $I$, for every fixed $E \in I$ and all sufficiently large $m$, periodic matrices of side $4^{20m}$ and heights $\eta_m \asymp 4^{-m}$ satisfy Theorem 8.2's certificate hypotheses with fixed positive constants $\alpha, \pi_0, B_1, p_-, p_+, q_*$ independent of $E, m$. Then almost surely $I \subset \sigma(H)$, $\chi_I(H)\ell^2 \subset \mathcal{H}_{\mathrm{ac}}(H)$, and $\mu_0(I) > 0$.*

*Proof.* Choose fixed $\theta, \epsilon$ as above. Translation invariance of the true finite law gives $\mathbb{E}\Phi_\theta(q_0^{(m)}(E + i\eta_m)) \ge c_1 > 0$; the construction need not be equivariant. Couple torus and infinite potentials on a root cube of side comparable to $L_m$, then sever its boundary bonds. Exponential conjugation changes nearest-neighbor hopping by $O_d(a\eta)$ in norm; the scalar-damped inverse has norm at most $\eta^{-1}$. A Neumann series for small $a$ gives $|G_{xy}(E + i\eta)| \le C_d\eta^{-1}e^{-c_d\eta|x-y|_1}$, also for cut operators, using bounded truncated weights in infinite volume. Resolvent identity across $O_d(L_m^{d-1})$ crossing bonds yields

$$|G_{00}^{(m)} - G_{00}| \le C_d L_m^{d-1}\eta_m^{-2}e^{-c_d\eta_m L_m} \longrightarrow 0. \tag{24}$$

Indeed $\eta_m L_m \asymp 4^{19m}$. The bounded Lipschitz function $\Phi_\theta$ transfers the expectation. The classical Herglotz boundary theorem for finite spectral measures gives finite complex diagonal boundary values almost everywhere. Fubini and bounded convergence give a deterministic full-measure subset of $I$ where $\mathbb{E}\Phi_\theta(q_0(E+i0)) \ge c_1$. This uses the classical boundary theorem, not a numerical limiting argument.

For each realization, every off-diagonal $G_{xy}$ is a difference of two Herglotz functions by polarization and has finite boundary values almost everywhere. It is not identically zero: the first nonzero $(x, y)$ term in $(H - z)^{-1} = -\sum_{n\ge 0} H^n z^{-n-1}$ is at $n = |x - y|_1$, and equals $(-1)^n$ times the positive number of shortest lattice paths. Such a difference is a quotient of bounded analytic functions. The boundary uniqueness theorem for a nonzero bounded analytic function implies $G_{xy}(E + i0) \ne 0$ almost everywhere. This input follows from Jensen's inequality for $-\log|f|$: a positive-measure set of zero boundary values would make a Poisson integral infinite although bounded above by $-\log|f(z_0)| < \infty$. Countability of site pairs and Fubini make these properties almost sure at each energy in a deterministic full-measure set.

Fix such an energy. The event of positivity at some site is translation invariant and has positive probability, so iid ergodicity makes it have probability one. For $x \ne y$ freeze all potentials except $s_x$. There is at most one admissible own-site value with $q_x > 0, q_y = 0$. Indeed at one such

value, positivity of the two-site imaginary matrix implies $a = G_{xy}$ is real; it is nonzero. With $g = G_{xx}$, changing the site by $h \neq 0$ gives

$$q_y(s_x + h) = \frac{h^2 a^2 \operatorname{Im} g}{|1 + hg|^2} > 0.$$

The denominator cannot vanish because $\operatorname{Im} g > 0$. The exceptional singleton and the inadmissible own-site set have zero probability under the density. Countably many pairs imply positivity at every site almost surely at that fixed energy.

Purity still requires a separate argument. Freeze the complementary configuration $\bar{\omega}$ at site $x$. The cavity formula is

$$G_{xx}^{(s)}(z) = \frac{1}{s - M_{\bar{\omega}}(z)}, \qquad M_{\bar{\omega}}(z) = z + \langle b, (H_{\bar{\omega}}^c - z)^{-1} b \rangle.$$

Fubini gives positive finite $q_x$ for almost every energy and $\rho(s)\,ds$-almost every $s$ for almost every complement. A finite real cavity boundary value would force imaginary part zero for every $s$ except one, a contradiction. Therefore $S_{\bar{\omega}} = \{E \in I : M_{\bar{\omega}}(E + i0) \text{ finite}, \ \operatorname{Im} M_{\bar{\omega}}(E + i0) > 0\}$ has full Lebesgue measure. It is Borel, complement-measurable, and on it every real own-site value has a finite diagonal boundary value.

A finite measure whose Poisson transform has finite limsup on a Borel set $S$ is absolutely continuous when restricted to $S$. To verify this, cover $S$ by $S_k = \{E \in S : \sup_{0<\eta<1/k} \operatorname{Im} g(E + i\eta) \leq k\}$. For $E \in S_k$, $r < 1/k$, one has $\mu((E - r, E + r)) \leq 2kr$. Cover a Lebesgue-null subset by open intervals with arbitrarily small total length, center an interval of twice each radius at a point of the subset, and apply this bound. It gives zero measure on that subset; countable union proves the assertion. Thus $\mu_{x,s}|_{S_{\bar{\omega}}} \ll dE$ for every $s$. For $B_{\bar{\omega}} = I \setminus S_{\bar{\omega}}$, own-site averaging gives

$$\int \rho(s) \mu_{x,s}(B_{\bar{\omega}})\, ds \leq \|\rho\|_\infty |B_{\bar{\omega}}| = 0.$$

Consequently $\mu_x|_I \ll dE$ almost surely. Intersect countably over sites; their projected basis vectors span a dense subspace of $\chi_I(H)\ell^2$, and $\mathcal{H}_{\mathrm{ac}}$ is closed. Finally, positivity of $q_0(E + i0)$ almost everywhere excludes any open spectral gap inside $I$ and forces $\mu_0(I) > 0$. Otherwise its Poisson transform would tend to zero at every interior point. □

Only the transfer is extended. The uniform-law binary digits, adaptive clearing eligibility and all-scale success estimate have not been proved for a different density. Even for the original law, our local certificates do not supply the premise. Positive mean density of states is weaker and can coexist with almost-sure pure point spectrum; positive finite-height imaginary parts also occur for finite pure-point matrices.

# 10 Verification, Limits, and Research Directions

## 10.1 What a defensible finite record contains

A state record must bind the site ordering, $A, T$, installed real potential, $E, t, r, p$, terminal-solve data, representation and metric, and retained history to the same version. It records full residuals including omitted components, active stability $tp_0$, Ward defect, relative form enclosure, $K$ and fourth-row margins, and exact range/rank evidence. A commit certifies its declared finite target

only. Failure retains the old numerical state while preserving every queried answer. An unknown quantity is not a passed test.

The local theorems permit all-charge energy checks rather than sampled vectors. The fixed-reference enclosure avoids automatically multiplying one tolerance per edit. An exact range basis can replace a zero-ridge limit test, and weighted radii inequalities make the local reset constants explicit. These are proved verification improvements, but no efficiency on the source's enormous scales is established.

### 10.2 Verification scope

The bounded author script completed successfully with 74 passing checks on eight five-dimensional fixtures and two negative tests, using seed 523. Its largest recorded identity defect was $1.14 \times 10^{-13}$. The script and output are stored under `_audit/M523-author*`; the final manifest binds their hashes to source and input hashes. Small prescribed matrices and ordinary double precision check residual signs, physical decomposition, dual correction, Ward defect, form enclosure, harmonic Schur defect, exact-range Gram bounds, and the negative small-singular-value example. These are algebraic consistency fixtures, not validated interval certificates or a simulation of the source construction.

An independent exact rational SymPy suite, `_audit/M523-parent-matrix-checks.py`, reports 13 passing groups: six noncommuting two-/three-dimensional Cayley fixtures, six corresponding normalized Ward/commutator/form fixtures with all PSD principal minors, and the scalar reset witness. These exact tests support the local algebra, not the adaptive history or infinite-volume premise. The negative rollback fixture confirms that a numerically restored state does not erase an observed sign. Final visual QA is recorded in the accompanying audit. No fixture is credited as proof of a universally quantified spectral statement.

The native editor opening was queued; native compilation returned a platform-directory failure, not a successful compile. Its exact response is retained separately in the audit. The disk PDF is exported by the existing `_tools/compile_report.py` using installed MiKTeX, without installation. Independent targeted report review found no consequential local error under the declared assumptions; that review did not certify the source's all-scale probability argument.

### 10.3 Current primary literature and comparisons

Public lookup on October 7, 2026 used only public mathematical topics and titles. Black, Drogin, and Hernández [2] study Gaussian disorder and a positive-height weak-coupling diffusion regime; v2 is dated November 6, 2025. Such estimates do not automatically supply a fixed positive-coupling zero-height certificate. Jakšić, Last, and Warzel [3], September 7, 2026, give a tree extended-state argument and formulate a lattice criterion using a two-site Poisson-transform Schur complement. The quantitative premise of a criterion still requires proof for the lattice model.

Becker and Oltman [4], September 17, 2026, announce absolutely continuous spectrum and exclusion of singular spectrum in specified hyperbolic-lattice regimes, with interval purity for specified laws. Their hyperbolic square-lattice product geometry is not the Euclidean lattice of M523. Aizenman and Warzel [5], Proposition 2.2, supply the historical conditional purity principle, not a transfer of tree delocalization to $\mathbb{Z}^d$. Tropp [6] supplies the primary matrix-martingale framework; source Section 5 proves the scalar-envelope version actually used. Public comparisons here are restricted to the checked metadata/abstracts and indicated primary passages, not independent verification of entire external papers. The limited search does not establish absence of other literature.

### 10.4 Pros, costs, and unmet dependencies

The positive result is an explicit finite audit invariant under appropriate physical-coordinate changes. It retains active damping, neutral compatibility, terminal sources, small singular directions, and the distinction between committed and provisional states. The weighted reset supplies a genuinely weaker local sufficient hypothesis with a verified witness, while the readout extension has a clearly delimited conditional boundary.

The costs remain substantial. Terminal pivots can be ill-conditioned; relative energy errors are stringent at low frequencies; complete residual envelopes may be dense; exact range equality is discontinuous under uncontrolled perturbations; and reset requires a verified inverse row bound. No representation name eliminates these requirements. No measured speedup, larger disorder threshold, mobility-edge location, transport exponent, or global delocalization result is claimed.

Remaining source dependencies include uniform Fourier initialization, all stopped conditional drift and variance estimates, full adaptive eligibility, spatial overlap and tails in assembly, witness entropy, and the infinite family of probability conditions for certificate existence. Targeted review of readout and reset does not certify that entire chain.

Future work should implement validated active/terminal residual actions and low-frequency form tests on modest deterministic fixtures, then investigate well-scaled full-range factorizations. An adaptive simulation needs an explicit query ledger demonstrating that rollback never reinstates freshness. A different site law requires a new centered exposure representation and reanalysis of clearing eligibility and moments. Effective thresholds and all-scale probability bounds require analysis, not finite extrapolation.

### 10.5 Contribution assessment

The realized assessment is C3/5, distinct from applicability level 5. The residual-to-form theorem, physical and terminal residual accounts, fixed-reference endpoint enclosure, and full-range test form a notable source-specific local verification extension. Explicit weighted reset constants and the witness beyond the sufficient $\|K\| < 1/2$ condition give an additional local gain. The Cayley factor improvement and conditional density extension support that contribution but are not historically new transform or global spectral results. No new all-scale certificate, validated large-scale implementation, or improved spectral theorem is obtained. This is an internal evidence-based assessment, not external peer review.

## A Read Scope and Evidence Locations

The source files and corpus paths in Sections 1 and 3 are hashed in the author input inventory. The catalogue row pin and policy are recorded separately from changing shared metadata. The source bibliography and figure were read; no separately included mathematical appendix exists in the supplied driver. Labels in Section 2 distinguish read dependencies from locally proved results.

Cooperative analytical provenance is recorded in `_audit/M523-parent-readout.txt`; it contains the sharp Cayley calculation, weighted reset derivation, and targeted source readout audit. Independent report-review findings, execution receipts, native failure, and final visual checks are retained in the local audit and final manifest. No source original, shared helper, catalogue, queue, progress file, or another report was modified by this author.

## References


[1] OpenAI, *Absolutely Continuous Spectrum for Weak-Disorder Anderson Models in Dimensions at Least Three*, September 23, 2026. Complete local source at the pin stated on the title page.

[2] A. Black, R. Drogin, F. Hernández, *Self-consistent equations and quantum diffusion for the Anderson model*, arXiv:2506.06468v2, November 6, 2025. https://arxiv.org/abs/2506.06468v2.

[3] V. Jakšić, Y. Last, S. Warzel, *A Note on Extended States on the Bethe Lattice*, arXiv:2609.06987v1, September 7, 2026. https://arxiv.org/abs/2609.06987v1.

[4] S. Becker, I. Oltman, *Disorder on the hyperbolic square lattice I: Anderson delocalization and absolutely continuous spectrum*, arXiv:2609.20798v1, September 17, 2026. https://arxiv.org/abs/2609.20798v1.

[5] M. Aizenman, S. Warzel, *Resonant delocalization for random Schrödinger operators on tree graphs*, J. Eur. Math. Soc. 15 (2013), 1167–1222, Proposition 2.2. https://ems.press/content/serial-article-files/31898.

[6] J. A. Tropp, *Freedman's inequality for matrix martingales*, Electron. Commun. Probab. 16 (2011), 262–270. https://tropp.caltech.edu/papers/Tro11-Freedmans-Inequality.pdf.

[7] G. Teschl, *Mathematical Methods in Quantum Mechanics*, AMS, 2009. Source Section 9 cites Theorem 3.7 and Corollary 3.25 for spectral calculus and Herglotz boundary values. These classical analytic inputs are used explicitly in Theorem 9.1.